# Bias-Aware Indic Spoken Language Identification: Task 1-3 Report

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## Abstract

This report covers all three project tasks for Indic spoken language identification on badrex/nnti-dataset-full. In Task 1, we reproduced baseline pipelines and compared pretrained backbones. In Task 2, we addressed speaker shortcut bias through data-centric augmentation and model-based adversarial training (DANN). In Task 3, we analyzed likely bias sources, confusion patterns, and representation behavior with t-SNE. Across complete runs, the strongest model in our branch is DANN over mHuBERT, improving validation accuracy from 0.4676 (untuned mHuBERT baseline) to 0.5576 and macro-F1 from 0.4142 to 0.5426 while preserving speaker-disjoint validation integrity.

## 1 Introduction

Spoken language identification (SLID) in low-resource, multi-speaker settings is vulnerable to shortcut learning: models may rely on speaker identity cues instead of language-specific cues. Our project is organized into three tasks: (1) reproduce and improve baseline performance, (2) mitigate speaker bias, and (3) analyze outcomes with quantitative diagnostics and visualization.

Our implementation is based on PyTorch (Paszke et al., 2019) and Hugging Face Transformers/Datasets (Wolf et al., 2020). For analysis tooling, we used scikit-learn (Pedregosa et al., 2011) and Matplotlib (Hunter, 2007). We evaluated pretrained speech encoders including MMS (Pratap et al., 2023), wav2vec 2.0 (Baevski et al., 2020), and mHuBERT (HuBERT family (Hsu et al., 2021)).

## 2 Data and Experimental Protocol

We use badrex/nnti-dataset-full (Badr et al., 2026), which provides predefined train/validation splits.

Key characteristics from diagnostics:

- 22 target languages.
- 8689 train samples, 3300 validation samples.
- 110 unique train speakers.
- 681 unique validation speakers.
- Speaker overlap between train and validation: 0.

Model selection is based on validation metrics with best-checkpoint loading enabled. We report accuracy, macro-F1, eval loss, per-language accuracy, confusion matrices, and speaker diagnostics.

### 2.1 Methods and Experimental Settings

To keep comparisons fair across Task 1/2/3 experiments, we kept the split protocol fixed and changed only the intended method components. Core controls were: fixed seed (42), same dataset split, same base backbone family (mHuBERT for final comparisons), and best-checkpoint selection rather than last-step reporting.

Task 1 experiments compare pretrained backbones and then a tuned mHuBERT reference. Task 2 experiments start from the tuned no-mitigation reference and introduce either (i) data-centric augmentation or (ii) adversarial speaker de-biasing. Task 3 uses run artifacts (metrics, confusion, speaker diagnostics, t-SNE outputs) for evidence-based analysis.

## 3 Task 1: Baseline Recreation and Improvement

### 3.1 Models and Results

Table 1 summarizes major Task 1 runs (validation split).

### 3.2 Task 1 Conclusion

In our environment, mHuBERT with tuning is the strongest Task 1 setting, improving over untuned

| Model / Run | Accuracy | Macro-F1 | Eval Loss |
|---|---|---|---|
| facebook-mms-300m | 0.2197 | 0.1892 | 2.7230 |
| wav2vec2-xls-r-300m | 0.0900 | 0.0486 | 3.0646 |
| mHuBERT (untuned) | 0.4676 | 0.4142 | 2.2854 |
| mHuBERT (tuned) | 0.5270 | 0.4748 | 2.1629 |

Table 1: Task 1 validation performance across backbone choices and tuning.

mHuBERT by +0.0594 accuracy and +0.0607 macro-F1. We use this tuned run as the no-mitigation reference for Task 2.

# 4 Task 2: Speaker-Bias Mitigation

## 4.1 Problem and Approaches

With limited speaker diversity per language in training, the model can reduce loss by exploiting speaker-correlated signals, creating a clear shortcut risk.

We evaluated:

- Data-centric mitigation (train-only augmentation): speed perturbation + waveform noise (Mitigation 1).
- Model-based mitigation: DANN with gradient reversal (Ganin et al., 2016).
- Intermediate ablations (Mitigation 2, 3A, 3B, 4) to study tradeoffs.

**Method details.** For data-centric mitigation, augmentation is applied only on train samples (not validation), with probabilistic speed/noise perturbation. For model-based mitigation, DANN uses a shared encoder with a language head and an adversarial speaker head through gradient reversal.

At encoder level, the optimization target is:

$$\mathcal{L}_{enc} = \mathcal{L}_{lang} - \lambda \mathcal{L}_{speaker}$$

which encourages language-discriminative but speaker-invariant features. In our final DANN run, we used `speaker_loss_weight`=0.1 and `grl_lambda`=1.0 with scheduling.

## 4.2 Mitigation Results

| Run | Acc | Macro-F1 | Loss |
|---|---|---|---|
| Tuned ref (no mitigation) | 0.527 | 0.475 | 2.163 |
| Mitigation 1 (data-centric) | 0.539 | 0.487 | 2.150 |
| Mitigation 2 | 0.528 | 0.483 | 2.139 |
| Mitigation 3A | 0.498 | 0.456 | 2.188 |
| Mitigation 3B | 0.528 | 0.482 | 2.146 |
| Mitigation 4 | 0.512 | 0.466 | 2.213 |
| DANN (final improved) | **0.558** | **0.543** | **1.881** |

Table 2: Task 2 mitigation results (validation split).

| Run | ΔAcc | ΔF1 | Train hrs |
|---|---|---|---|
| Tuned ref (no mitigation) | 0.000 | 0.000 | 5.51 |
| Mitigation 1 | +0.012 | +0.012 | 6.65 |
| Mitigation 2 | +0.001 | +0.008 | 6.17 |
| Mitigation 3A | -0.029 | -0.019 | 9.24 |
| Mitigation 3B | +0.001 | +0.007 | 9.18 |
| Mitigation 4 | -0.015 | -0.008 | 5.99 |
| DANN (final improved) | **+0.031** | **+0.068** | 8.18 |

Table 3: Ablation summary relative to tuned reference.

## 4.3 Extent of Success

Relative to untuned baseline mHuBERT, DANN improves by +0.0900 accuracy and +0.1284 macro-F1. Relative to Mitigation 1, DANN still improves by +0.0188 accuracy and +0.0558 macro-F1.

Speaker-oriented evidence:

- Unseen-speaker validation accuracy: 0.4676 (untuned) → 0.5576 (DANN).
- Train/validation speaker overlap remains 0 (no leakage warning).
- Key bias-sensitive confusions reduced, e.g., `sindhi`→`punjabi`: 79 → 14.

These results indicate meaningful mitigation of speaker-related shortcuts, but not complete removal of speaker information.

## 4.4 Ablation and Efficiency Summary

To make the mitigation decision explicit, Table 3 reports deltas relative to the tuned no-mitigation reference.

Interpretation: Mitigation 1 gives a stable data-centric gain; Mitigation 2 and 3B are near-tied but below Mitigation 1; Mitigation 3A regresses; Mitigation 4 underperforms Mitigation 1; DANN provides the strongest final improvement, with higher training cost but a better accuracy/F1 tradeoff.

# 5 Task 3: Analysis

## 5.1 Source of Bias and Learning Difficulty

The data regime (few train speakers per language, fully unseen validation speakers) creates strong shortcut pressure. Early optimization may latch onto speaker timbre/prosody, while weaker classes remain under-separated. This pattern appears in near-zero class performance for the untuned baseline (e.g., maithili, konkani) and concentrated confusion clusters.

| Pair (true→pred) | Baseline | DANN | Δ |
|---|---|---|---|
| konkani→marathi | 106 | 74 | -32 |
| sindhi→punjabi | 79 | 14 | -65 |
| hindi→urdu | 60 | 39 | -21 |
| santali→bengali | 54 | 10 | -44 |
| dogri→punjabi | 53 | 44 | -9 |
| bodo→manipuri | 39 | 45 | +6 |

Table 4: Confusion-pair count changes from untuned baseline to DANN. Negative Δ is an improvement (fewer errors).

### 5.2 Confusion Analysis

Most frequent confusions include:

- `konkani→marathi`
- `sindhi→punjabi`
- `hindi→urdu`

Likely drivers include linguistic/acoustic proximity and imbalance in class difficulty. DANN reduces several dominant pairs but does not eliminate all structural confusion.

### 5.3 Representation Analysis (t-SNE)

We applied t-SNE (van der Maaten and Hinton, 2008) to last-layer validation embeddings for the improved run. Baseline/tuned/M1 t-SNE export was not reproducible from current local artifacts because those run folders do not contain full model/preprocessor checkpoint files.

For DANN:

- Silhouette (language): 0.0047
- Silhouette (speaker top-k): -0.0992
- kNN speaker purity@5: 0.3433

Interpretation: speaker identity appears reduced but is still partially encoded.

### 5.4 Additional Insightful Analyses

To avoid single-metric conclusions, we added the following quantitative analyses:

- Class-balance profile across runs (mean/median/min/max per-language accuracy).
- Weak-class recovery index (baseline weak set: hindi, konkani, maithili, santali, sindhi).
- Error concentration (top-5 confusion mass ratio).
- Best-step ratio (best step / final step) to show late-epoch plateau and justify best-checkpoint reporting.

These analyses are documented in `reports/task3_analysis.md` and `reports/tables/`.

## 6 Reproducibility Notes

We report the exact run IDs and artifacts used in analysis (`reports/artifacts_used.md`). Core reproducibility settings are:

- Seed: 42.
- Training epochs: 3.
- Mac profile for final mitigation runs: batch size 1, gradient accumulation 4, max duration 7 s, max grad norm 1.0, evaluation/save every 100 steps.
- Best-checkpoint loading enabled; mitigation runs selected with macro-F1 criterion.
- Speaker diagnostics and confusion matrices generated for every final compared run.

For Mitigation 3B and Mitigation 4, summary metrics in this report are taken from archived project reports in `reports/` because the corresponding full run directories are not present in the current branch snapshot. These controls reduce accidental variance and support a fair mitigation comparison.

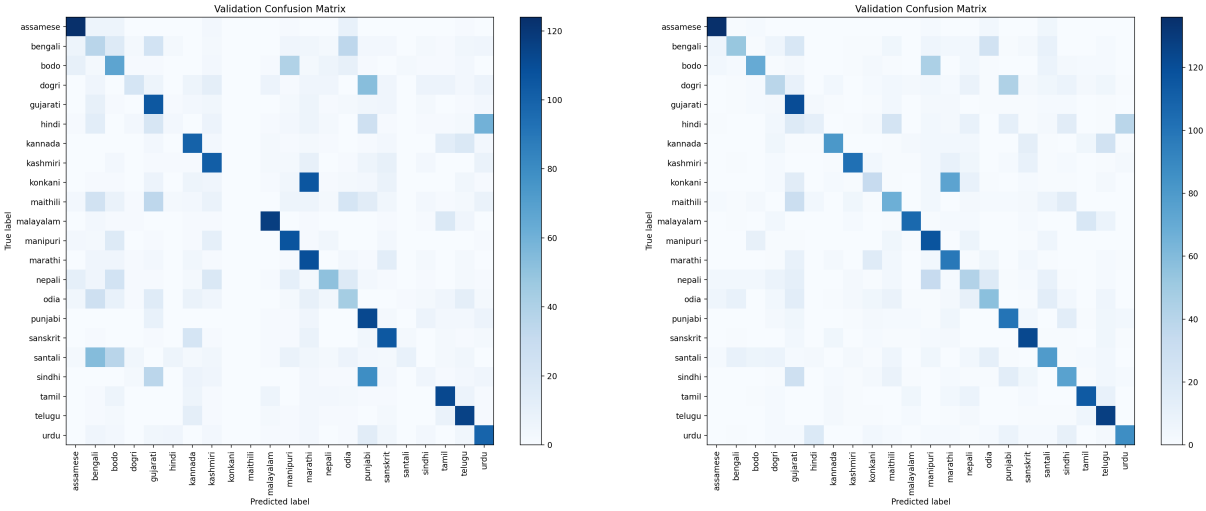


Figure 1: Validation confusion matrices: untuned baseline (left) vs DANN improved (right).

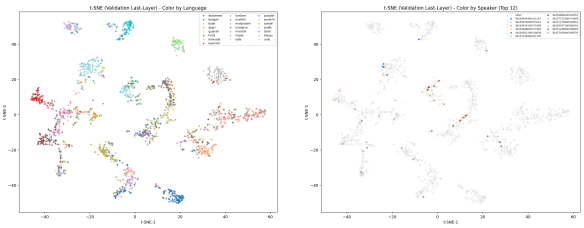

Figure 2: DANN t-SNE on validation embeddings: language coloring (left) and speaker coloring (right).

## 7 Conclusion

Across all three tasks, mHuBERT is the strongest backbone in our environment. Data-centric mitigation improves balanced performance, and DANN yields the best final model. The final DANN run achieves 0.5576 validation accuracy and 0.5426 macro-F1 with clean speaker-disjoint diagnostics. Remaining challenges are concentrated in a small set of linguistically close language pairs.

## Limitations

First, the strongest t-SNE comparison currently uses improved-run embeddings, while baseline/tuned/M1 t-SNE export is limited by missing checkpoint artifacts in those run folders. Second, some persistent confusions (e.g., konkani-marathi) remain unresolved and may require additional data balancing or task-specific objectives. Third, compute constraints limit the number of long-run and combined-mitigation experiments.

## Ethics Statement

This work focuses on technical mitigation of dataset bias and does not deploy a user-facing system. We explicitly monitored speaker overlap and unseen-speaker behavior to reduce leakage-driven claims. Because language recognition systems can be misused for profiling, we report calibrated conclusions and avoid claims of complete bias removal.

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